

Dear students,

The final exam for the "MTM3651 Statistics" course will be held on Thursday, January 11, 2024, at 09:00.

The exam will be a hybrid exam containing also classical and multiple-choice questions.

Statistics tables and the following formulas will be distributed during the exam. (The formulas below will be given in the exam question paper, and the formulas given on the second page will be distributed to you as a separate paper). Bring a simple calculator with you to the exam. The use of mobile phones and the sharing of calculators will not be permitted.

Good luck.

$\chi_0^2 = \sum_{i=1}^k \frac{(O_i - E_i)^2}{E_i}$	$\chi_0^2 = \sum_{i=1}^r \sum_{j=1}^c \frac{(O_{ij} - E_{ij})^2}{E_{ij}}$	Relative Efficiency $= \frac{Var(\theta_2)}{Var(\theta_1)}$
$\sqrt{\frac{N-n}{N-1}}$	$F_0 = \frac{MSB}{MSW}$	D. C. = $\frac{\sigma}{\mu} 100$
$SSB = \sum_{i=1}^k n_i (\bar{x}_i - \bar{x})^2$	$SSW = \sum_{i=1}^k \sum_{j=1}^{n_i} (x_{ij} - \bar{x}_i)^2$	$SST = \sum_{i=1}^k \sum_{j=1}^{n_i} (x_{ij} - \bar{x})^2$

STATISTICS FORMULAS

$S_k^2 = \frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{n_1 + n_2 - 2}$ $v = n_1 + n_2 - 2$	$v = \frac{\left[\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2} \right]^2}{\left(\frac{S_1^2}{n_1} \right)^2 \frac{1}{n_1 - 1} + \left(\frac{S_2^2}{n_2} \right)^2 \frac{1}{n_2 - 1}}$	$P_{\text{C.K.}} = \frac{3(\mu - \text{Me})}{\sigma}$
$\text{Mo} = L + \frac{\Delta_1}{\Delta_1 + \Delta_2} \cdot c$	$S = \frac{M_3}{\sigma^3}, M_k = E[(X - \mu)^k]$	$K = \frac{M_4}{\sigma^4} - 3$
$Q_i = L_i + \left(\frac{\frac{n_i}{4} - (\sum f)_i}{f_i} \right) \cdot c$	$B_{\text{C.K.}} = \frac{Q_3 + Q_1 - 2\text{Me}}{Q_3 - Q_1}$	$\frac{(n-1)S^2}{\sigma^2} = \chi_{n-1}^2$
$r = \frac{n \sum XY - (\sum X)(\sum Y)}{\sqrt{[n \sum X^2 - (\sum X)^2][n \sum Y^2 - (\sum Y)^2]}}$		$\frac{S_1^2 / \sigma_1^2}{S_2^2 / \sigma_2^2} = F_{n_1 - 1, n_2 - 1}$
$R^2 = \frac{\sum (\hat{Y} - \bar{Y})^2}{\sum (Y - \bar{Y})^2}$	$S_r = \sqrt{\frac{1 - r^2}{n - 2}}$	$\text{Var}(\hat{\theta}) = \frac{1}{n \cdot E \left[\left(\frac{\partial \ln f(X)}{\partial \theta} \right)^2 \right]}$
$t_{n-2} = \frac{r - \rho}{S_r}$	$S^2 = \frac{\sum (Y - \hat{Y})^2}{n - 2} = \frac{\sum e^2}{n - 2}$	$S_{\widehat{\mu_{Y_k}}} = \sqrt{\frac{1}{n} + \frac{(X_k - \bar{X})^2}{\sum X^2 - n\bar{X}^2}} S$
$E(b_0) = \beta_0 \quad E(b_1) = \beta_1$	$\text{Var}(b_0) = \frac{\sum X^2}{n \sum X^2 - (\sum X)^2} S^2$	$\text{Var}(b_1) = \frac{1}{[\sum X^2 - (\sum X)^2 / n]} S^2$
$t_{n-2} = \frac{b_0 - \beta_0}{\sqrt{\text{Var}(b_0)}}$	$t_{n-2} = t = \frac{b_1 - \beta_1}{\sqrt{\text{Var}(b_1)}}$	$t_{n-2} = t = \frac{\hat{Y}_k - Y_k}{S_{\hat{Y}_k}}$
$\hat{p}_w = \frac{n_1 \hat{p}_1 + n_2 \hat{p}_2}{n_1 + n_2},$	$P \left(\frac{\mu_0 - \mu_1}{\sigma / \sqrt{n}} \mid -z_\alpha \leq z_\beta \right) = \beta$	
$P \left(\frac{\mu_0 - \mu_1}{\sigma / \sqrt{n}} - z_{\alpha/2} \leq z_\beta \leq \frac{\mu_0 - \mu_1}{\sigma / \sqrt{n}} + z_{\alpha/2} \right) = \beta$	$P \left(z_\beta \leq \frac{\mu_0 - \mu_1}{\sigma / \sqrt{n}} + z_\alpha \right) = \beta$	
$n = \frac{(z_\beta + z_{\alpha/2})^2 \sigma^2}{(\mu_0 - \mu_1)^2}$	$n = \frac{(z_\beta + z_\alpha)^2 \sigma^2}{(\mu_0 - \mu_1)^2}$	
<u>Sum of Total Squares</u> (STS): $\sum (Y - \bar{Y})^2 = \sum Y^2 - (\sum Y)^2 / n$ <u>Sum of Regression Squares</u> (SRS): $\sum (\hat{Y} - \bar{Y})^2 = b_1^2 [\sum X^2 - (\sum X)^2 / n]$ <u>Sum of Error Squares</u> (SES): $\sum (Y - \hat{Y})^2 = STS - SRS$		$k = \sqrt{n}$ $k = 1 + 3, 3 \log n$ $k = 5 \log n$